

ment relations and of controlling labour standards via formal agreements on wages and other non-commercial features such as gender and racial discrimination.

- *Power sharing*—an ethos of co-operation and trust replaces the adversarial relations endemic to command-and-control regulation. Partnerships involve mutually beneficial sharing of responsibility, knowledge or risk, negotiating the differences that were otherwise litigated. For neo-conservatives, power sharing makes regulation less rigid.

This grammar of meanings becomes useful when the partners come (and they do, as a rule) from different sectors of administration, politics or professions and become transitional or continuous players in the evolving game. Because of its ideological roots, the word *partnership* conveys strong normative premises that are central in the trends of welfare state reforms. PPP players who overlook (or are unaware of) this fact may experience painful disappointments.

The mathematical theory of games

For several decades the mathematical theory of games has been described and treated in the literature as a self-consistent scientific discipline. Tutorial examples illustrate the concepts and theorems. Sophisticated theoretical models and business applications prove the power of the approach (Shubik, 1982; Aliprantis and Chakrabarti, 2000). Some of the most significant applications are never published because of their strategic or economic importance. Regular courses in engineering, and many of those in economics, however, do not traditionally incorporate this field of knowledge. The participants of PPP games are not usually acquainted with the basic concepts and relationships of the mathematical theory of games, except for the few coming from financing and banking. The latter think that having this knowledge is a question of competence; all partners have to decide whether they need it and use it, or not, when bargaining about their interests.

Instead of referring the reader to introductory textbooks, this section of the paper informally reviews the fundamental concepts and theorems necessary to understand the benefits of the game theory perspective. It is intended for the reader who used to play without the knowledge of game theory and is interested in a better understanding of what s/he is doing. Those acquainted with the theory may simply skip this section or enjoy the pleasure of checking its phraseology suited to common language and engineering attitude.⁴

Games consist of two or more participants: the players. Each player has a set of alternative choices governed by the rules of the game; s/he can select a strategy (a plan of her/his sequential choices or moves) to arrive at the outcome of the game. The possible choices and moves correspond to the rules of the game. The actual outcome depends on the players' strategies; for each possible outcome each player gets a pay-off. It is assumed that all players behave rationally: they seek their best possible outcome. No social or human motivations are considered except those quantified and introduced in the form of rules or, though implicitly, coded in the utility functions expressing the preferences of the players. The selfishness of a player, for instance, is coded as indifference to the desires of others (Shubik, 1975).

Both individuals and groups can participate in games as players. Delegates having additional and particular utility functions not necessarily identical with that of their dispatcher may represent some of them. The attitudes of the players (preferences regarding the utility of different outcomes, *risk* sensitivity, readiness to make decisions under *uncertainty*, etc.) influence the decision they consider as rational. Different classes of games can be defined:

- two-person games differ significantly from n -person ones ($3 \leq n$);
- the players may or may not know the strategies of each other;
- the players can be informed *completely* or *incompletely*, *perfectly* or *imperfectly* about the rules and possible outcomes of the game they are playing;
- the strategy sets of the players may be small or large, finite or infinite (even empty);
- the sum of the pay-offs may be fixed (even zero) or may depend on the players' performance and co-operation;
- one-step, repeated, *sequential*, finite and infinite games exist with different possibilities to learn about, adapt, etc.;
- the rules of the game can be 'rigid' (well defined in advance) or 'free' (in this case some rules can be determined or interpreted only as the game proceeds).

Simple and complex games are described in detail in the available literature; many of them have been constructed or applied with reference to the transportation industry and project financing. Appendix B presents several examples.

'Non-cooperativeness' is the natural atmosphere of finite two-person zero-sum games where one player wins exactly what the other loses. In most games with more than two players the possibility exists of creating coalitions. Interest in a specific outcome, desire for a higher pay-off, fear of loss, altruism, moral incentives, etc. can induce co-operation. The players may evaluate the social environment of the game. Some players' decisions can be influenced by *side-payments* given to compensate a member who joined the coalition with a sub-optimal individual pay-off. Players with a different amount and reliability of information often enter sub-games.

In sequential n -person games there might be *dummy* players who do not have the chance to influence decisions made by other players though they may have a stake in the outcome. The *strategic* dummy is strictly constricted by the rules and has no choice in her/his actions.

Additional assumptions and equilibrium principles are established to identify *solutions*: more specifically, to determine the outcome that corresponds to the rules and strategies of the players in an optimal sense. Sophisticated further concepts are introduced to model the more complex gaming patterns (e.g. sequential games played by coalitions of players with imperfect information). Refined assumptions and theorems govern the more delicate games in which the players may have mixed strategies to cope with the random behaviour of their partners. Because of the non-linear character of the governing relationships, the existence and uniqueness of the solution is a question of particular importance.

The concepts, assumptions, governing principles and relationships applied in the mathematical theory of games may appear unusual to many professionals but are neither more nor less difficult than those applied in the natural sciences, particularly in mechanics or thermodynamics. Some principles are not analogous but identical. Saddle-point theorems are well known in both fields though interpreted in different ways. The same computational methods based on minmax principles (such as the Kuhn–Tucker theorem), and programming techniques (linear, quadratic, etc.) are applied in mechanics and in game theory alike.

Psychology

Games—as in transaction analysis (Berne, 1964, 1972)—are sets of complementary, ulterior, often repetitive moves with a lure or snare and a concealed motivation progressing to a well-defined, predictable psychological outcome. Such psychological

games can be distinguished from other types of procedures (transactions, operations, missions, etc.) by their ulterior quality and built-in pay-off. Psychological games involve contest, they are basically dishonest and their outcomes are dramatic.

The simplest two-person game described by Berne is played by the initiator (agent) and the respondent (mark). The initiator is doing one thing while he is really doing something else. He has a con to hook into a gimmick and controls a switch in order to get the pay-offs of the game (usually feelings). The respondent has a gimmick of weakness (fear, greed, etc.). Given this basic mechanism, four moves constitute the game:

- the initiator (agent) sets the con—his response places the respondent (mark) at a weak position;
- the initiator pulls the con to get hold of the gimmick;
- the respondent gets confused;
- both players collect their pay-offs.

Psychological games are classified into life games, marital games, party games, sexual games, underworld games, and consulting room games. To date, no direct analogies between psychological and PPP games have been possible. In turn, the motivations of the players, the structures and other pattern-related components of PPP games seem to be worth scrutinizing from the perspective of transaction analysis.

The theory of institutions

Social and economic institutions (such as international credit markets, regional governments or bus queues) constitute a framework for accomplishing collective actions. In the broad sense they mean '*the rules of the game*' (North, 1990). Their performance depends on the presence or absence of what is known as 'social capital'. This concept can be defined in several ways (Hardin, 1999):

- a set of different entities consisting of some aspects of social structures and facilitating some actions (Coleman);
- a substance incorporating the influence of past actions by peers and others into preferences in an individual's social network (Becker);
- a set of social connections and the attendant norms and trusts that enable participants to act together more effectively (Putnam);
- the ability of people to work together for common purposes in groups and organizations (Fukuyama).

In what follows we apply the interpretation given by Putnam. In this perspective, social attitude related to co-operative or uncooperative behaviour 'depends on the broader social context within which any particular *game is played*'. Trust, norms of reciprocity and networks of civic engagement constitute a substantial stock of social capital.

Like the more conventional forms of capital (such as physical or human capital), social capital is productive. It makes it possible to achieve certain goals that are otherwise unattainable. In dilemmas of collective action it helps the individual to circumvent problems of imperfect information. Communities can leverage their social capital to expand available credit facilities and improve the efficiency of markets. Though productive, social capital is also a moral good. Its forms increase or strengthen with use and exhaust or diminish with disuse. It is ordinarily a public good, unlike conventional capital. That is why 'like all public goods, social capital tends to be undervalued and undersupplied by private agents' (Putnam, 1993, p. 170).

The emergence and accumulation of social capital is not a process of altruistic effort. Trust is based on the expectation that the partner will choose the option of common interest. It lowers transaction costs and facilitates collective action. The norm of reciprocity serves to reconcile self-interest and solidarity. Networks of civic engagement increase the potential cost to a defector in any individual transaction; foster robust norms of reciprocity; facilitate communication and improve the flow of information; and embody past success at collaboration.

Basically, two ultimate alternative equilibrium states of social capital accumulation have been described. Virtuous circles result in stable and high levels of co-operation, trust, reciprocity and civic engagement, while the denial of co-operation, distrust, shirking, and isolation intensify one another in a vicious circle. Either alternative may evolve and, once attained, tends to be self-reinforcing (Putnam, 1993, p. 177).

The basic strategy of the latter state ('always defect') is stable for the reasons well explicated with respect to *the prisoner's dilemma of game theory*. Actors in this social equilibrium may wish to get to the co-operative state but to do so is beyond their power. The co-operative state prefers the strategy of the 'mutual-aid game', as described in game theory: 'cooperate with people who cooperate with you (or who cooperate with people like you) and don't be the first to defect' (Sugden, 1986). Economic development does not imply growing social capital. In post-industrial societies impersonal co-operation is crucial, whereas 'the returns on opportunism, cheating, and shirking rise' (North, 1990). In this perspective, state and market do not constitute alternative mechanisms for solving social problems: both operate more efficiently in civic settings with a higher level of social capital.

Identification of public–private partnership patterns from the various perspectives

Both common language and more rigorous professional analysis suggest that it makes sense to interpret PPP schemes as games. All concepts and processes occurring can be described properly in the language of 'gaming'. There is no danger of misunderstanding. The seriousness of the subject is not threatened: other disciplines have already canonized the vocabulary.

Gaming can be understood as an activity or a set of interactions involving players with a complex character, particular strategy, behaviour, interest, etc. The actions of the players may extend to the technological, political, economic, social and legal fields. At this level of generality the perspectives of several sciences can be applied to PPP. Other, more specified senses (connected with training, teaching, experimental, etc. purposes and applications) are not excluded by this interpretation and can be specified when introduced. In some cases even metaphorical or allegorical wording is both possible and expressive.

The consistent application of the gaming terminology offers a coherent set of familiar expressions such as rules, outcomes, coalitions, etc. More sophisticated concepts used in the game theories (such as risk aversion, lure, etc.) can be identified and used in PPP schemes as well. In most games with public participation (e.g. in those played for community development) the pay-offs are complex. Various notables or groups give only a crude verbal description of their goals, and their delegates will play the game as participants. Road infrastructure PPP games are often long and the instructions given to the players may change depending on the results of other ongoing games.

Both experience and perspectives show that the idealized pure market and public mechanisms cannot be identified in existing or workable public road management models. Most PPP schemes described in the literature can be identified as *sequential*

n-person games with imperfect and incomplete information. On a large scale there are at least three phases to be distinguished in road development procedures: soliciting, contracting and implementing (Morgan, pers. commun.). Obviously, all phases consist of several steps, depending on the technical conditions, on the relationships among the players, etc.

The selection of the participants, setting of the rules, selection of the players and timing of their involvement are often part of the PPP game. Some players enter sub-games: developers hire designers and contractors, financing institutions establish a sponsoring consortium, public authorities set up a specific public company, etc. The force of law outlines some of the rules; others may be bargained and accepted by the players. Some games can be repeated; the same players often enter slightly different games and make use of their prior experience and knowledge. There are opportunities in the game to think about goals and learn lessons.

Individuals and groups may have different 'ego-states'. They may address each other in different ways, depending on their interests (Ferrigno, 1990). They initiate stereotypical calculated games and leave them when the partners turn out to be well acquainted with the pattern. Some PPP players can be pressed or manipulated this way: their attitude is altered before the issue is clearly discussed.

Some risks identified in PPP schemes can be better understood in a more general context. The theory of institutions defines and analyses the risks that are generated by the institutional schemes themselves. For example, the 'expedient' estimation of toll revenues may help to launch a project but jeopardizes the financial stability of the operational phase; this risk did not exist until the toll-financed PPP approach was accepted.

There are significant applications of game theory to investigate road-infrastructure-related issues. The Nilsson model (Nilsson, 1996) highlights the importance of information in public service decision making. Its most relevant messages are:

- all parties involved in decision making must have access to the same information about relevant parameters—if this is the case then the importance of the institutional structure decreases and private and public performance may prove to be equally efficient;
- asymmetric and/or imperfect information and *ex ante* uncertainty about any preferences of the administration outside public interest undermine public-sector integrity;
- instead of vague declarations about the responsibility to the public, the administration has to be clearly instructed and obliged to legitimize its decisions using advanced assessment tools;
- public media monitoring, external audits and occasional reviews accomplished by independent civil entities are essential to ensure that decisions are congruent with declared policy;
- if the administration is 'accountable' and the risk of strong private preferences is low, then privatization may be expensive and public-sector production is preferred.

Governmental bodies having different responsibilities for transport and for fiscal balance usually play another game that crosses the PPP game. The government may be interested in leveraging (actually, transposing) its burdens, even at the expense of higher costs in the long run. Different departments assess timing constraints and external benefits differently. The attitude of the political administration is always ambivalent: it is aware that the burdens taken with a PPP in the long run can be significantly higher than in the case of a budget-financed project, but the leadership may be interested in responding to short-term political demands.

The road infrastructure PPP literature highlights the necessary redistribution of the roles to be played by participants but suggestions as how to do this are scarce and simplistic. New roles are particularly difficult to identify for the administration where culture and social capital, multiple responsibilities towards the public and the existing (sometimes contradictory) legislative environment do not necessarily support such changes. Private partners often become tired of uncertainty: ‘... the contractor said it knew how to deal with a public client and how to deal with a private client but it could not deal with an entity that could not decide what it was’ (Lammie, 1997).

Institutional actions can strongly influence the environment and the rules of a PPP game. In some cases alternative public financing techniques can emerge and compete. For instance, local jurisdictions may introduce local option taxes to raise revenues for the public financing of particular transportation improvements. A recent survey (Beale *et al.*, 1996), investigating the results of referenda frequently required before such legislative decisions, showed that the following factors may provide solid public support for this technique:

- an identified critical need for transportation improvements and design of a tax suitable to meet those needs;
- a thorough planning process that involves and informs the electorate and produces a realistic expenditure plan;
- detailed earmarking of funds in the expenditure plan;
- a suitable level, duration and type of tax;
- effective public–private leadership, provision of information, and promotion of the expenditure plan and the tax to finance it;
- fulfilment of promises, to build support for future referenda;
- thoughtful drafting of the ballot language and a suitable strategy for implementing the referendum;
- distribution of tax burdens, tax revenues and benefits that are perceived as fair;
- neutralization or integration of special interests;
- neutralization of anti-tax sentiments;
- a reasonably strong local economy.

It is worth noticing that some of these criteria are identical to those connected with successful PPP deliveries, while the others are considered as crucial to the accumulation of social capital.

Summary and conjectures

The concepts, phraseology and aspects of the general theories concerning games and the applications of these theories in various fields of everyday life prove to be useful in the analysis and practical implementation of PPP. The conceptual identification of positions, motivations and further characteristics is not only possible but helps all partners to better understand the structure, rules and roles of the procedure they are involved in. Applying the gaming perspective in PPP issues seems to enhance ‘*fair play*’.

The main message of this paper is that *the gaming perspective makes it possible to clarify some cloudy, uncertain or misinterpreted situations, positions or actions experienced in PPP games*. Many concepts, theorems, results and methods developed and used successfully in other fields can be transferred and applied to PPP schemes, particularly in road transportation infrastructure projects. Some of these are simple definitions or interpretations, others offer a deeper insight and help to structure the pattern of a PPP. Most of them need

further scrutiny, but can be accepted at least as plausible conjectures at the given level of analysis.

By definition, in most PPP games there are players (such as politicians and public servants) whose *utility scale* is plausibly not linear in money terms, since their world is not that of the market economy. Experienced private players reckon with these sensitivities and from the very beginning of the game try to accommodate themselves to the social environment and to the anticipated time constraints of their public partners.

Politicians are usually inclined to focus their attention on the next election. For them, a PPP project is worth supporting to the extent that it can be used to gain voters. Their utility function may have singularities both in technology, public budget and time dimensions. Politicians often enter PPP games with easily identifiable pay-off functions and strategies. Other players usually believe that strong political support is vital and helps to control many risks. This is true, but timing is a key issue and political involvement is controversial, as it may increase external risks. The engagement (if it was crucial with regard to project implementation) of a political group defeated at an election may block the best project for years. This kind of *risk sensitivity* is not identical to the general political uncertainty in politically unstable countries and may occur in developed democracies as well.

Decision makers and managers in public administration usually have sub-linear utility functions with a strong aversion to risks. They seldom prefer high-risk schemes, do not want 'to lay their careers on the line to exploit more efficient trade-offs', and try to avoid setting precedents, even if the corresponding pay-off is high. Other players who would like to see a public partner with a linear utility function can relax their *risk aversion* in different ways. The fears or dislikes of any partner can be moderated with side-payments. However, in most situations the declared rules of the game do not allow for such modifications, and such actions imply additional risks not necessarily listed in PPP schemes.

Psychological games, by definition, include a *lure* and some of them are, in fact, played in a commercial environment. Their generalization to enterprises or groups is more or less straightforward. In PPP games, lures may naturally appear since the private entities entering the project do not waive their strategies for dealing with each other simply because they have a public partner. These lures, however, are laid in the sub-games played by the private partners. They are of minor interest when we consider the PPP game itself. Here, the specific lure attracting the public partner is of primary importance.

In the transportation sector the public-related specific lure is the promise of a better infrastructure with 'almost no extra burden on taxpayers', a facility that improves conditions for the community as a whole. In fact, the promise (declared either by a politician, a local official or a private executive) is trustworthy. Furthermore, significant groups may discover particular external benefits offered by the project for clever users. However, the real question is who pays for what. In the vast majority of PPP projects (if not in all of them) the *general public* is not told that it is taken as the *ultimate owner* and user who, either directly or indirectly, will pay all the costs of the facility. The only decision variable is how to allocate costs across groups and over time.

Individuals concerned in road transportation infrastructure issues (and their representatives or delegates, as players) have particular '*ego-states*' more or less corresponding to Berne's classification, as shown in Table 1.

The same person approaches the same question differently, depending on her/his attitude. The individual is able to respond at the level s/he was addressed, but general cultural traditions, norms or the public and private partners' actions can distort the response.

Table 1

Berne structure	PPP structure
Parent	Taxpayer
Adult	Citizen
Child	Consumer

As in psychological games, the lured *mark* (or *patsy*) is not perfectly innocent and often behaves as a deliberate victim. The general public (in other words, the average taxpayer or resident) likes the idea of a free lunch. A highway user may be so much inclined to ignore the externalities connected with her/his prompt benefit as to forget about the asset-related obligations of the community.

Further lures may appear in many PPP games. For instance, powerful partners do not disclose the character or the rules of the game they want to play. In the first phase they may conduct more than one game simultaneously and change or modify the line depending on the partners' moves.

PPP allows for a rich variety of participants. Players may enter of their own accord, they can be invited, or become involved by virtue of their position. By definition, the community is a player. Its role, however, is split up among public authorities, agencies, organizations and interest groups. The 'ultimate public', the community of citizens, is usually a *dummy* player (i.e. it has a role but does not have a chance to make decisions). When the community is the ultimate owner of the transportation facility, it plays the role of the *strategic dummy* in many PPP schemes. Its *pay-off* is to bear all burdens leveraged, bridged, or transferred upon it with skilful financing techniques in the long run that were not taken by identified beneficiaries.

Financially viable PPP projects almost never demand a strategic dummy of this kind. At the other extreme, however, there are cases where the revenues collected from the beneficiaries never add up to the costs. Then, a strategic dummy is needed. It may or may not be told that the investors finance the costs; and all public burdens (if any) do not have a direct impact on the budget or on the taxation level, respectively. This may be true *linguistically*. But the content of PPP contracts determining in strict legal terms how the public partner will take over all burdens of private investments (and their fair or not so fair profits) is seldom disclosed and is usually declared a business secret. The strategic dummy may wonder if costs (in properly discounted and comparable terms) may be higher than in the case of direct public financing but is never offered the chance to see the difference in clear balance sheets.

Whether or not the community is a mark, a strategic dummy or a properly represented player in a PPP game, *depends* on many conditions. That is why PPP cannot be judged 'in general' in terms of common welfare and ethics. It is worth noting, however, that the psychological element of a lure and the role of the dummy often appear coupled; the lure assumes that at least one of the players has to be uninformed about its position and its possible strategies and it has to remain constrained in its moves.

The proliferation of PPP models helps to commercialize those public services where the traditional position of users and providers has become indeterminate in the welfare state and is not transparent enough to ensure *the fair allocation* of their costs and benefits in a public game. However, if the additional costs involved in the partnership are financed by a dummy player (such as the taxpayer or the citizen), the social capital of the general public suffers a loss.

Common-pool theories and their corresponding game-theoretical models (see Appendix B) have special importance for the interpretation of PPP games. It is important, however, to see the significant difference between *common pools* supplied by Nature (such as a hunting ground or a meadow) and the goods (such as a road or a fish-pond) *created* and owned by a community.

The social capital of the community—a ‘soft’ value defined in the theory of institutions—is embodied as ‘hardware’ in some common goods such as *public road* assets. Financing the public road infrastructure is an issue of creating and/or maintaining *social capital*. Some groups are deeply interested in consumption; others are devoted to accumulation. Fair groups seek a balance between consumption and accumulation. In road infrastructure games it makes sense to identify consumer groups with regard to their readiness to consume or accumulate social capital. The overlap between the sets of criteria to successfully deliver a PPP project, to maintain and increase social capital in a community and to manage a *pro* referendum for an optional tax to finance a public transport infrastructure facility is impressive.

Trends, expectations and areas of further research

PPP is expected to remain an important instrument of transportation policy implementation, political power enforcement and financial capital investment. Its proliferation is in harmony with the continuing deconstruction of public administration and with the popularity of mortgage plans in which individuals, communities and generations leave debts to their descendants. Alternative financing techniques are important because they give governments at any level new options to access available revenues and to smooth and leverage them to meet the expenditures over the time period until public resources become available. Social sciences may clarify the qualitative and quantitative consequences of this trend.

More disciplined public asset accounting (Kadlec and McNeil, 2001) and elucidation of the differences between services and assets can always enhance more skilled public participation in PPP schemes. In general, the better understanding of PPP games can reduce the losses and costs, even if the outcomes are not governed by consensual moral or free-market principles. The legal environment of PPP activity has to be improved. In particular, amendments are needed to regulate the evaluation phase to achieve as much transparency and competition as possible.

The reallocation of roles in the transportation infrastructure development industry is a challenge for education. Engineers have to expand their knowledge (Pethtel and Scharle, 2001). Pure professional technical skills are becoming undervalued and inadequate to influence the outcomes governed by other disciplines’ paradigms.

Further analysis of delivered PPP projects (whether successful or failed) is needed to develop an adequate terminology, description techniques and methodology for supporting the delivery of future projects. Case studies of PPP games can be developed for training and teaching purposes and used to enhance the better understanding of partnership issues.

The experience and lessons learned in European PPP practice during the last two decades have to be analysed to identify several particular business-culture-dependent alternatives and trends that are different from those present in the USA.

Specific interdisciplinary research may be focused on the:

- possible conflicts between PPP delivery and standard controlling mechanisms (they